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Roku Resource Monitor session file specification

Developers can follow this specification to build a custom resource monitoring tool that analyzes the memory, CPU, and SceneGraph node usage metrics collected with the Roku Resource Monitor (/docs/developer-program/dev-tools/resource-monitor.md). This specification describes the data types and structure of a monitoring session file (.JSON) that has been exported from the Roku Resource Monitor.

> This specification has been updated to support Roku Resource Monitor 4.0.
>
> The v1 version of this spec and the session files generated by the previous versions of the Roku Resource Monitor (v1.0 and v1.1) have been deprecated.

File name

The name of an exported Roku Resource Monitor file has the following syntax: `<channel-name>_<device-name>_<unix-timestamp>.json`.

Data types

The fields in the monitoring file may be one of the following data types:

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Data type	Definition
string	Includes alphanumeric and special characters
number	Includes integer and fractional numbers.
bool	A true/false value.

File structure

A monitoring session file contains the following objects (listed in order):

Object	Description
device	The name, model, serial number, and OS version of the test device.
channel	The ID, name, and version of the test app.
metadata	The unique ID, timestamp, and version of the monitoring file.
session	A stream of multiple entries of the following types: • static. The background and foreground DRAM usage limits (for supported devices only; this field is set to "null" on unsupported devices). • live. The resource usage statistics. This contains arrays of objects for the following metrics: channel_system_memory_usage, channel_cpu_usage, channel_graphics_memory_usage, channel_graph_metrics, graphics_rendering_frame_rate, sgrendezvous and fw_beacons.

Device information

The device object contains the following fields that provide information about the test device used for the monitoring session:

Name	Type	Description	Example
device_name	string	The device-friendly name.	"Roku Express"
model_number	string	The device model number.	"3930X"
serial_number	string	The device serial number.	"X00400CS3EMP"
software_version	string	The software version installed on the device.	"23.0.0"

Name	Type	Description	Example
software_build	number	The software version build installed on the device.	991

Channel information

The channel object contains the following fields that provide information about the test app used for the monitoring session:

Name	Type	Description	Example
id	string	The unique identifier of the app	"581251" or "dev"
version	string	The app version	"1.1.2"
name	string	The app name	"Roku Developers"

File metadata

The metadata object contains the following fields that provide information about the JSON file generated for the monitoring session:

Name	Type	Description	Example
uuid	string	A randomly-generated Version 4 unique user ID (UUID).	cd7cbbeb-c4a7-46dc-ab9f-fdeb90523481
created_at	number	A unix timestamp in milliseconds (13-digit) when the file was created.	1657721930416
started_at	number	A unix timestamp in milliseconds (13-digit) when the monitoring session was started.	1657721930416
version	number	The file version.	4

Session information

The session object contains the static memory usage limits of the test app and the live resource usage statistics generated during the monitoring session.

Static object

The static object contains the following persistent data:

Name	Type	Unit	Description
foreground_limit	number/null	Bytes	The maximum amount of DRAM that may be used by the app when browsing and selecting content in the app UI and during playback. If an app exceeds the limit displayed in the app system memory usage pane while running in the foreground, the Roku OS terminates the app. The foreground limit varies by device, and it is subject to change. On devices that do not support per-app memory limits, the Roku Resource Monitor displays "N/A". The Roku OS, however, does terminate apps running in the foreground when specific system memory levels are reached on those devices. The foreground limit may be decreased in the near future; therefore, apps should consume only 75% of the displayed limit. The reduced limit will be enforced by the Roku OS in the near future.

Name	Type	Unit	Description
background_limit	number/null	Bytes	The maximum amount of DRAM that may be used by the app while running in the background (this limit is also applicable for apps that have integrated Instant Resume). If an app exceeds the limit displayed in the app system memory usage pane while running in the background, the Roku OS terminates the app. The Roku OS also terminates apps running in the background when specific system memory levels are reached. On devices that do not support per-app memory limits, the Roku Resource Monitor displays "N/A". The Roku OS, however, does terminate apps running in the background when specific system memory levels are reached on those devices. The background limit may be decreased in the near future; therefore, apps should consume a maximum of 100 MB of DRAM while running in the background. The reduced limit will be enforced by the Roku OS in the near future. See Data management (/docs/developer-program/performance-guide/data-management.md) and Memory management (/docs/developer-program/performance-guide/memory-management.md) for best practices on allocating resources.

Live

The live object contains the following objects for the resource usage statistics generated during the monitoring session:

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Name	Type	Description
bs_objects	array	Array of points that include BrightScripted objects

Name	Type	Unit	Description
timestamp	number	Bytes	Unix timestam obtained
groups	array	N/A	Array of object

Name	T
Name	s
Id	s
Objects count	n
Objects	a
Objects memory	n

Name	Type	Description
channel_system_memory_usage	array	Array of points for the following system memory us

Name	Type	Unit	Description
timestamp	number	Bytes	Unix timestam obtained
used	number	Bytes	Number/null. A and swap) by th
resident	number	Bytes	Number/null. A app (sum of sha
swap	number	Bytes	Number/null. A
anon	number	Bytes	Number/null. A the app
file	number	Bytes	Number/null. A app
shared	number	Bytes	Number/null. A

channel_cpu_usage	array	Array of points for the following CPU usage metrics

Name	Type	Unit	Description
timestamp	number	Bytes	Unix timesta obtained
total	number	Percent	Number/null. user and kerr
user	number	Percent	Number/null. app
sys	number	Percent	Number/null. the app

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Name	Type	Description
channel_graphics_memory_usage	array	Array of points for the following graphics memory u

Name	Type	Unit	Description
timestamp	number	Bytes	Unix timestam obtained
texture	number	Bytes	Number/null. A app
system	number	Bytes	Number/null. A app
bitmaps	array	N/A	Array of object

Name	Typ
bpp	nun
height	nun
width	nun
size	nun
name	stri

Name	Type	Description
channel_graph_metrics	array	Array of points for the following SceneGraph metri background):

Name	Type	Unit	Description
timestamp	number	Bytes	Unix timest was obtaine
total_nodes	number	Bytes	Number/nu
total_memory	number	Bytes	Number/nu
Nodes	array	-	Array of obj

Name
Name
Count
Memory

Name	Type	Description
graphics_rendering_frame_rate	array	Array of points for the following graphics rendering (foreground and background):

Name	Type	Unit	Description
timestamp	number	Bytes	Unix timestam obtained
fps	number	Count	Recent numbe

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Name	Type	Description
sgrendezvous	array	Array of points for the following rendezvous event (background):

Name	Type	Unit	Description
timestamp	number	Bytes	Unix timestam obtained.
total_time	number	Bytes	The total time sampling interv
drop_count	number	Bytes	The number of because the ev
count	number	Bytes	The number of interval
items	array	N/A	Array of object event informat

Name
Id
start
end
line_number
name

Name	Type	Description
fw_beacons	array	Array of app and media events (signal beacons) tha Unix timestamp in milliseconds (13-digit) marking v a number type provided in bytes. See Measuring ap program/performance-guide/measuring-channel-p these events. Channel events: • app_compile_initiate / app_compile_complete • app_load_initiate / app_load_complete • app_splash_initiate / app_splash_complete • app_resume_initiate / app_resume_complete • app_suspend_initiate / app_suspend_complete • app_exit_initiate / app_exit_complete Media events: • vod_start_initiate / vod_start_complete • epg_launch_initiate / epg_launch_complete • live_channel_change_initiate / live_channel_ch
registry_usage	array	Array of points for the channel registry entries.

Name	Type	Unit	Description
Timestamp	number	Bytes	Unix timestam obtained.
Used	number	Bytes	Number/null. T the device.
Sections	array	-	Array of object

Name	Typ
Name	stri
Items	arra

Monitoring file example

Click here (https://devtools.web.roku.com/rrm/rrm-file-spec-4-1-example.json) to download a sample RRM 4.1 monitoring file.

Change log

- v4.1 (June 16, 2025):
  - Updated bs_objects and channel_graph_metrics objects.
- v4.0 (March 12, 2025):
  - Updated bs_objects field.
  - Updated sample file.
- v3.1 (September 24, 2024):
  - Added fields for tracking BrightScript memory allocation..
- v3.0 (April 11, 2024):
  - Added objects for tracking BrightScript memory allocation.
- v2.0 (March 28, 2023):
  - Added objects for tracking rendezvous events, device registry usage, and graphics frame rendering rate.
  - Added objects for tracking app and media events.
- v1.0 (September 11, 2022):
  - Initial release.
  - Includes objects for tracking system memory, graphics memory, CPU, and SceneGraph node usage.

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